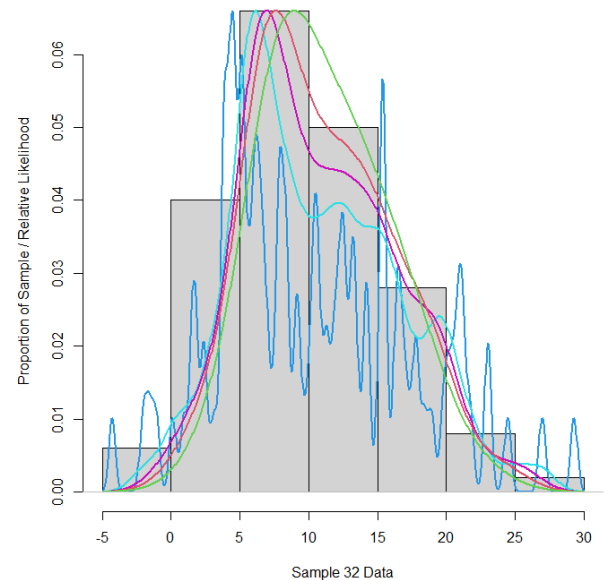


# Assignment 3

## Exercise 3.2

For this exercise we pick the gaussian kernel function because it is the correct one for most of the occasions. For the bandwidth to start with 0.2 (blue), this bandwidth had too much detail. The next bandwidth with less detail was 1 (light blue) with still too much noise in the line. The peak is moving to the mean and median but is still improvable. The line with the bandwidth 1.5 (pink) is nice but the peak is still a bit too low. The line with bandwidth 2 (red) looks good with the peak and its tails are according to the data symmetry. 3 (green) the peak is even better than by two, the line shows a normal distribution what the data is not. This can be seen in the plot on the right.

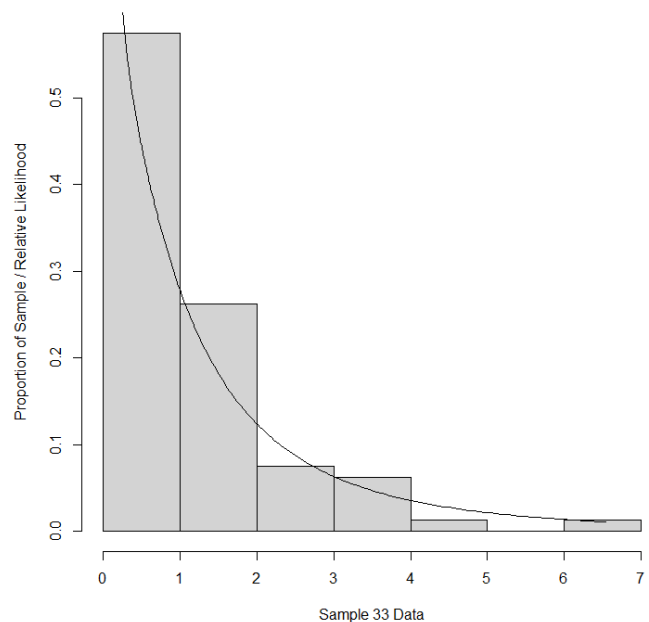
Histogram and Kernel Density Estimate of Sample 32



## Exercise 3.3

We want our density estimate to be zero for non-positive data because all the observations are positive. Hence, we log-transform *sample33* data. Then we find corresponding optimal bandwidth with the given function *h\_opt*. Then we use the function *density* with the gaussian kernel and the optimal bandwidth to determine a suitable kernel density estimate for the log-transformed data. We choose the gaussian kernel by default as the choice of kernel is not crucial. Then we use the given formula in the assignment to back-transform the density estimate of the log-transformed data to obtain the density estimate of the original sample. Notice the histogram of the original data and its corresponding density estimate.

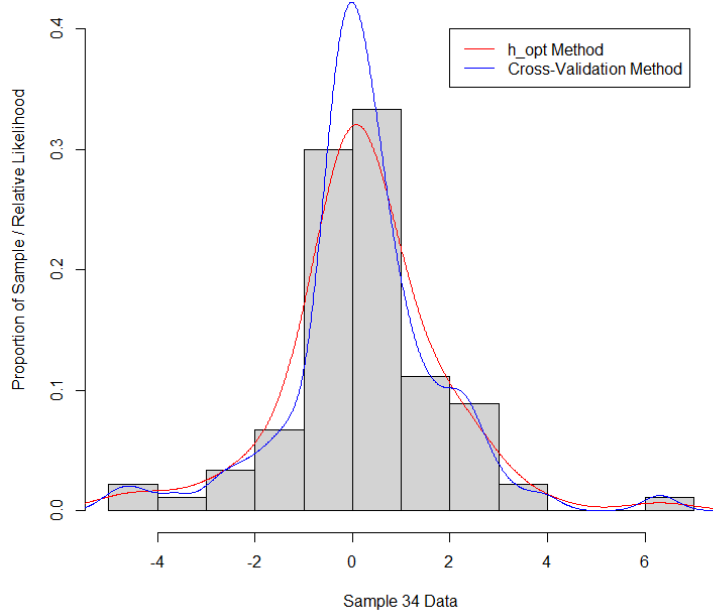
Histogram and Kernel Density Estimate of Sample 33



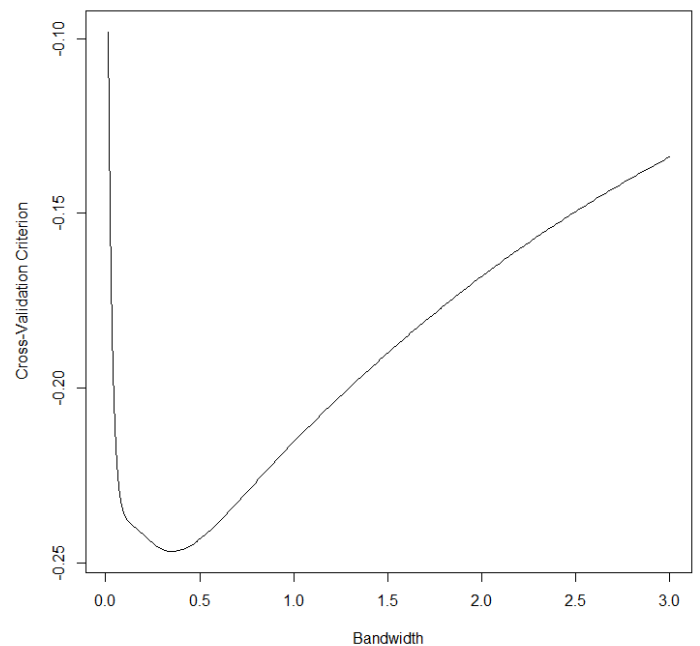
### Exercise 3.4

We use the gaussian kernel in both cases. In the first case, the function  $h\_opt$  gives an optimal bandwidth value of 0.69. In the second case, minimizing the function  $CV$  for various bandwidth values between 0 and 3 gives an optimal bandwidth value of 0.35. This can be seen in the plot on the right.

**Histogram and Kernel Density Estimates of Sample 34**

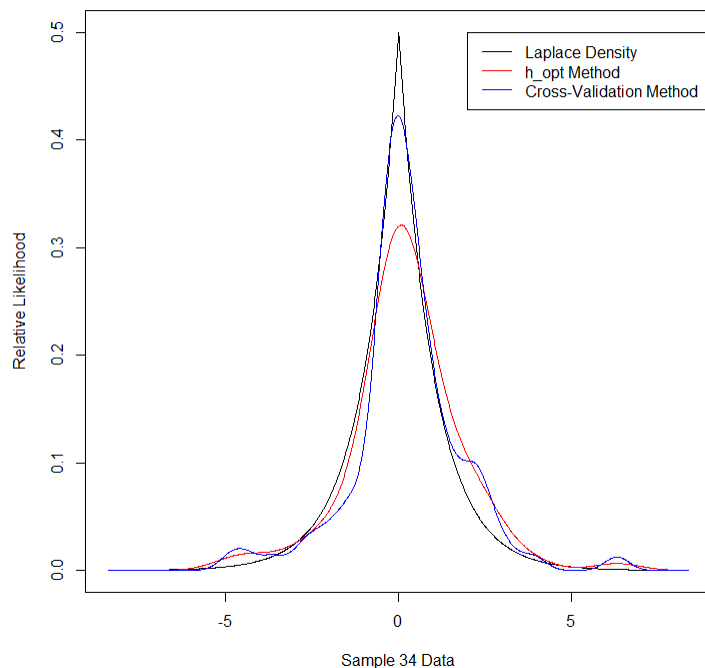


**Bandwidth Minimizing The Cross-Validation Criterion**



The two different bandwidth values give rise to two different kernel density estimates shown in the histogram on the left.

**Laplace Density and Its Kernel Density Estimates**



We construct the double exponential density, the Laplace density, by gluing together two exponential densities with rate 1 and multiplying them by 0.5 to obtain area of 1. The plot on the left shows the true Laplace density and its kernel density estimates.

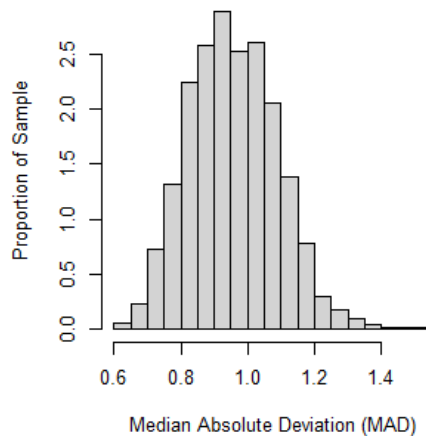
The plot shows that the density obtained by the CV method better matches the true density around the point 0 but it is rather “wobbly” near and at its tails. The density obtained by the  $h\_opt$  function matches the smoothness of the Laplace density except for the “sharp” summit near the point 0. This makes sense as the greater bandwidth is a smoothing factor. If we were to decide which density to pick, then we would pick the density obtained from the  $h\_opt$  function because the density obtained by the CV method is too “wobbly” in our opinion.

## Exercise 3.5

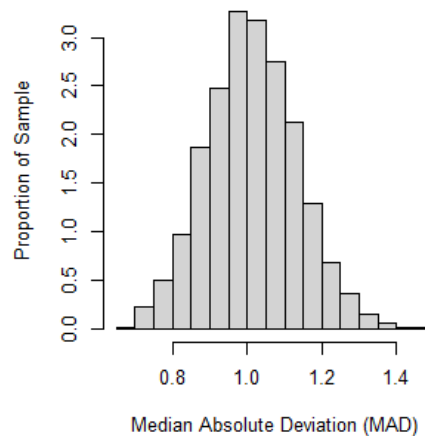
### Part d.

Visually, the histograms are similarly bell-shaped and have similar tails. Notice that the overall mass on right of  $MAD = 1$  is greater than the overall mass on left of  $MAD = 1$  in the histogram of realizations of the true MAD statistic. This holds for the parametrically bootstrapped MAD sample but not for the empirically bootstrapped MAD sample. Therefore, we opt for the parametric bootstrap method.

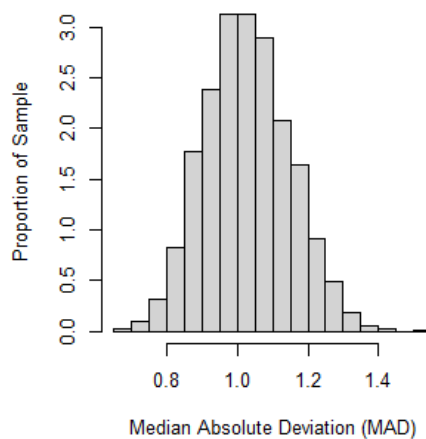
**Histogram of Empirically Bootstrapped MAD**



**Histogram of Parametrically Bootstrapped MAD**



**Histogram of Realization of True MAD**



### Part e.

|                      | variance of the median absolute deviation statistic |
|----------------------|-----------------------------------------------------|
| Empirical bootstrap  | 0.0173                                              |
| Parametric bootstrap | 0.0149                                              |
| Realization          | 0.0164                                              |

The Empirical is closed but too high and Parametric is a little too low. If they are combined, it can probably be used as a strong Bootstrap method.

## Appendix

```
setwd("C:/Users/david/OneDrive/VU/Statistical Data Analysis/Assignment 3")
setwd("~/Documents/BBA3/SDA")
source("sample33.txt")
source("sample34.txt")
source("t-sample.txt")
source('functions_Ch4.txt')
source('functions_Ch5.txt')
```

### # Exercise 3.2

```
data32 <- scan("sample32.txt")
summary(data32)
par(mfrow=c(1,1))
hist(data32, breaks = 5,freq = FALSE,
      main = "Histogram and Kernel Density Estimate of Sample 32",
      xlab = "Sample 32 Data",
      ylab = "Proportion of Sample / Relative Likelihood")
par(new = T)
plot(density(data32, kernel = "gaussian" ,bw = 0.2), lwd = 2,xlab = "", ylab = "", main = "", sub = "",col =
4,axes = FALSE)
par(new = T)
plot(density(data32, kernel = "gaussian" ,bw = 1), lwd = 2,xlab = "", ylab = "", main = "", sub = "",col =
5,axes = FALSE)
par(new = T)
plot(density(data32, kernel = "gaussian" ,bw = 1.5), lwd = 2,xlab = "", ylab = "", main = "", sub = "",col =
6,axes = FALSE)
par(new = T)
plot(density(data32, kernel = "gaussian" ,bw = 2), lwd = 2,xlab = "", ylab = "", main = "", sub = "",col =
2,axes = FALSE)
par(new = T)
plot(density(data32, kernel = "gaussian" ,bw = 3), lwd = 2,xlab = "", ylab = "", main = "", sub = "",col =
3,axes = FALSE)
```

### # Exercise 3.3

```
data33 <- scan("sample33.txt")
hist(data33, prob=T, main = "Histogram and Kernel Density Estimate of Sample 33",
      xlab = "Sample 33 Data",
      ylab = "Proportion of Sample / Relative Likelihood")
y = log(data33)
yrange <- seq(min(y), max(y), length.out=512)
lines(exp(yrange), density(y, bw = h_opt(y),
      from=min(yrange), to=max(yrange))$y/exp(yrange))
```

### # Exercise 3.4

```

data34 <- scan("sample34.txt")

# h_opt method
h1 = h_opt(data34)

# cross-validation criterion method
hrange = seq(0.015, 3, length.out=300)
cv_criterion = lapply(hrange, CV, data34, "g")
plot(hrange, cv_criterion, type="l",
     main = "Bandwidth Minimizing The Cross-Validation Criterion",
     xlab = "Bandwidth",
     ylab = "Cross-Validation Criterion")
h2 = hrange[which.min(cv_criterion)]

# kernel density estimates
x_half = seq(0, max(data34) + 3*h1, length.out=600)
x = append(- rev(x_half)[1:599], x_half)
hist(data34, prob=T, ylim=c(0,0.45),
     main = "Histogram and Kernel Density Estimates of Sample 34",
     xlab = "Sample 34 Data",
     ylab = "Proportion of Sample / Relative Likelihood")
lines(x, density(data34, bw = h1, from = min(x), to = max(x), n = 1199)$y,
     col = "red")
lines(x, density(data34, bw = h2, from = min(x), to = max(x), n = 1199)$y,
     col = "blue")
legend(2, 0.4, legend=c("h_opt Method", "Cross-Validation Method"),
     col=c("red", "blue"), lty=1:1)

# double exponential comparison
doublexp = append(0.5 * rev(dexp(x_half))[1:599], 0.5 * dexp(x_half))
plot(x, doublexp, type="l",
     main = "Laplace Density and Its Kernel Density Estimates",
     xlab = "Sample 34 Data",
     ylab = "Relative Likelihood")
lines(x, density(data34, bw = h1, from = min(x), to = max(x), n = 1199)$y,
     col = "red")
lines(x, density(data34, bw = h2, from = min(x), to = max(x), n = 1199)$y,
     col = "blue")
legend(2, 0.5, col=c("black", "red", "blue"), lty=1:1,
     legend=c("Laplace Density", "h_opt Method", "Cross-Validation Method"))

# Exercise 3.5

# Part a.
tsample <- scan("t-sample.txt")
mad_sample = mad(tsample)

```

```

# Part b.
set.seed(20220341)
mad_empBS = bootstrap(tsample, mad, B = 2000)

# Part c.
s2 = var(tsample)
k = 2 * s2 / (s2 - 1)
mad_parBS = replicate(2000, mad(rt(100, k)))

# Part d.
par(mfrow=c(2,2))
hist(mad_empBS, prob = T, breaks = 20,
     main = "Histogram of Empirically Bootstrapped MAD",
     xlab = "Median Absolute Deviation (MAD)",
     ylab = "Proportion of Sample")
hist(mad_parBS, prob = T, breaks = 20,
     main = "Histogram of Parametrically Bootstrapped MAD",
     xlab = "Median Absolute Deviation (MAD)",
     ylab = "Proportion of Sample")
mad_realizations = replicate(2000, mad(rt(100, 10)))
hist(mad_realizations, prob = T, breaks = 20,
     main = "Histogram of Realization of True MAD",
     xlab = "Median Absolute Deviation (MAD)",
     ylab = "Proportion of Sample")

#Part e.
var_empBS = var(mad_empBS)
var_parBS = var(mad_parBS)
var_realizations = var(mad_realizations)

stud_no <- c(2669137,2671011)

mylist <- list(stud_no = stud_no, mad_sample = mad_sample, mad_empBS = mad_empBS,
mad_parBS = mad_parBS, mad_realizations = mad_realizations, var_empBS = var_empBS, var_parBS
= var_parBS, var_realizations = var_realizations)
save(mylist, file="myfile3_39.RData")

```